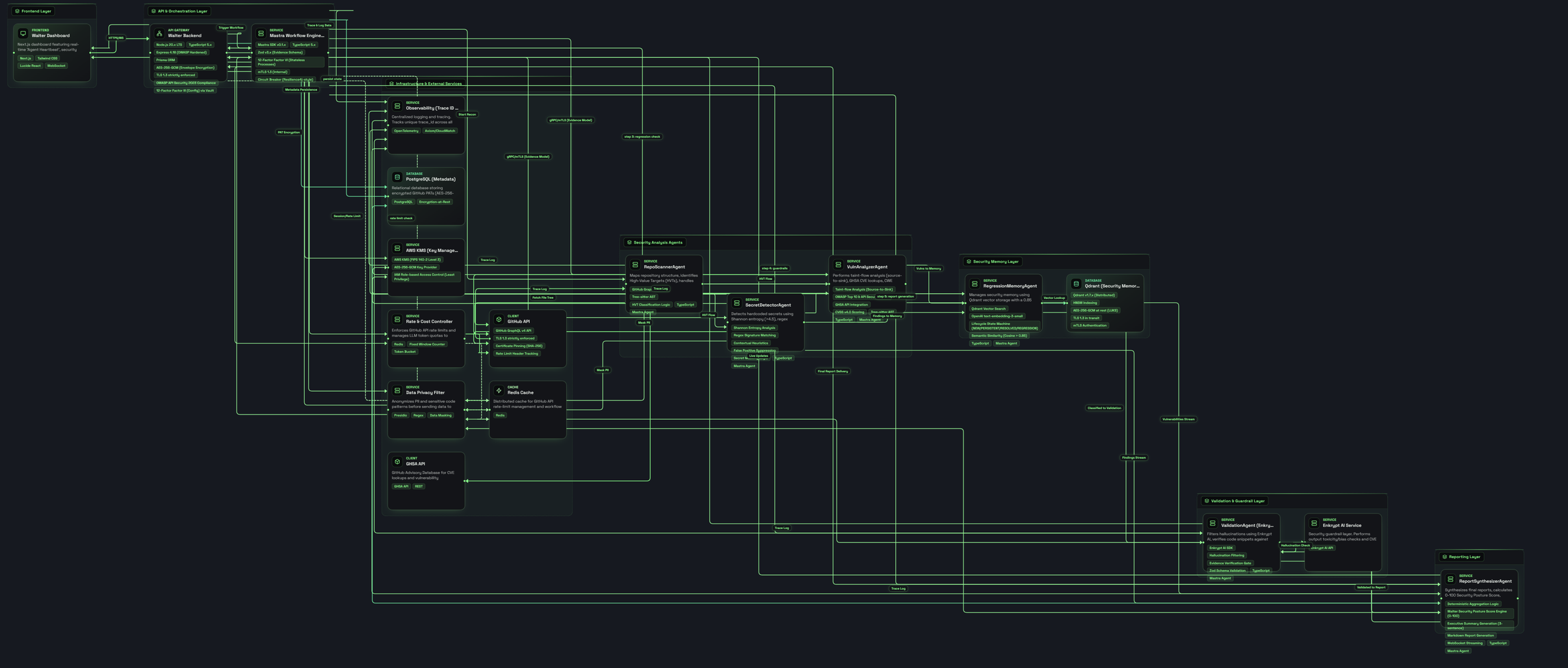
Frontend Lager
FRONTEND
Walter Dashboard
Next.js dashboard featuring real-time 'Agent Heartbeat', security
Next.js
Tailwind CSS
Lucide React
WebSocket
HTTPS/WS
API & Orchestration Lager
API GATEWAY
Walter Backend
Node.js 20.x LTS
TypeScript 5.x
Express 4.18 (OWASP Hardened)
Prisma ORM
AES-256-GCM (Envelope Encryption)
TLS 1.3 strictly enforced
OWASP API Security 2023 Compliance
12-Factor Factor III (Config) via Vault
Trigger Workflow
SERVICE
Mastra Workflow Engine...
Mastra SDK v0.1.x
TypeScript 5.x
Zod v3.x (Evidence Schema)
12-Factor Factor VI (Stateless Processes)
mTLS 1.3 (Internal)
Circuit Breaker (Resilience4j style)
Trace & Log Data
persist state
Metadata Persistance
Infrastructure & External Services
SERVICE
Observability (Trace ID ...
Centralized logging and tracing. Tracks unique trace_id across all
OpenTelemetry
Axiom/CloudWatch
Start Recon
gRPC/mTLS (Evidence Model)
step 3: regression check
gRPC/mTLS (Evidence Model)
PAT Encryption
DATABASE
PostgreSQL (Metadata)
Relational database storing encrypted GitHub PATs (AES-256-
PostgreSQL
Encryption-at-Rest
rate limit check
Session/Rate Limit
SERVICE
AWS KMS (Key Manage...
AWS KMS (FIPS 140-2 Level 3)
AES-256-GCM Key Provider
IAM Role-based Access Control (Least Privilege)
Trace Log
Security Analysis Agents
SERVICE
RepoScannerAgent
Maps repository structure, identifies High-Value Targets (HVTs), handles
GitHub Grap
Trace Log
Tree-sitter AST
HVT Classification Logic
TypeScript
Mastra Agent
Trace Log
Fetch File Tree
HVT Flow
Mask PII
step 4: guardrails
HVT Flow
SERVICE
VulnAnalyzerAgent
Performs taint-flow analysis (source-to-sink), GHSA CVE lookups, CWE
Taint-flow Analysis (Source-to-Sink)
OWASP Top 10 & API Secu
step 5: report generation
GHSA API Integration
CVSS v4.0 Scoring
Tree-sitter AST
TypeScript
Mastra Agent
Findings to Memory
Vulns to Memory
Security Memory Lager
SERVICE
RegressionMemoryAgent
Manages security memory using Qdrant vector storage with a 0.85
Qdrant Vector Search
OpenAI text-embedding-3-small
Lifecycle State Machine (NEW/PERSISTENT/RESOLVED/REGRESSION)
Semantic Similarity (Cosine > 0.85)
TypeScript
Mastra Agent
Vector Lookup
DATABASE
Qdrant (Security Memor...
Qdrant v1.7.x (Distributed)
HNSW Indexing
AES-256-GCM at rest (LUKS)
TLS 1.3 in transit
mTLS Authentication
SERVICE
Rate & Cost Controller
Enforces GitHub API rate limits and manages LLM token quotas to
Redis
Fixed Window Counter
Token Bucket
CLIENT
GitHub API
GitHub GraphQL v4 API
TLS 1.3 strictly enforced
Certificate Pinning (SHA-256)
Rate Limit Header Tracking
SERVICE
SecretDetectorAgent
Detects hardcoded secrets using Shannon entropy (>4.5), regex
Shannon Entropy Analysis
Regex Signature Matching
Contextual Heuristics
False Positive Suppression
Secret Sc
Live Updates
TypeScript
Mastra Agent
Mask PII
Final Report Delivery
SERVICE
Data Privacy Filter
Anonymizes PII and sensitive code patterns before sending data to
Presidio
Regex
Data Masking
CACHE
Redis Cache
Distributed cache for GitHub API rate-limit management and workflow
Redis
CLIENT
GHSA API
GitHub Advisory Database for CVE lookups and vulnerability
GHSA API
REST
Classified to Validation
Vulnerabilities Stream
Findings Stream
Validation & Guardrail Layer
SERVICE
ValidationAgent (Enkryp...
Filters hallucinations using Enkrypt AI, verifies code snippets against
Enkrypt AI SDK
Hallucination Filtering
Evidence Verification Gate
Zod Schema Validation
TypeScript
Mastra Agent
Hallucination Check
SERVICE
Enkrypt AI Service
Security guardrail layer. Performs output toxicity/bias checks and CVE
Enkrypt AI API
Trace Log
Trace Log
Validated to Report
Reporting Layer
SERVICE
ReportSynthesizerAgent
Synthesizes final reports, calculates 0-100 Security Posture Score,
Deterministic Aggregation Logic
Walter Security Posture Score Engine (0-100)
Executive Summary Generation (3-sentence)
Markdown Report Generation
WebSocket Streaming
TypeScript
Mastra Agent